Case Study

DS 4002 – Spring 2026

Submission format:

- **Upload link to GitHub repository**

What am I going to do? In this case study, you will create two models: an image-only model for image classification, and an image+metadata model for image classification. A starter script has been built for you with a pretrained image extractor model (ResNet18) and a logistic regression image classifier. Following completion of these two models, you should evaluate each performance using accuracy scores, precision scores, recall scores, and F1-scores.
All of this will be submitted electronically via a link to a github repository.

Formatting	● Repository – A GitHub repo containing all materials o A README.md file o A LICENSE.md file (use MIT as default) o A SCRIPTS folder o A DATA folder, including all data used o An OUTPUT
README.md	● Goal: This file serves as an orientation to everyone who comes to your repository, it should enable them to get their bearings. ● Use markdown headers to divide content. ● Make an H2 (##) section explaining the contents of the repository ● Section 1: Software and platform section o The type(s) of software you used for the project. o The names of any add-on packages that need to be installed with the software. o The platform (e.g., Windows, Mac, or Linux) you used. ● Section 2: A Map of your documentation. In this section, you should provide an outline or tree illustrating the hierarchy of folders and subfolders contained in your Project Folder, and listing the files stored in each folder or subfolder. ● Section 3: Instructions for reproducing your results. In this section, you should give explicit step-by-step instructions to reproduce the Results of your study. These instructions should be written in straightforward plain English, but they must be concise, but

	detailed and precise enough, to make it possible for an interested user to reproduce your results without much difficulty.
LICENSE.md	- Goal: This file explains to a visitor the terms under which they may use and cite your repository. - Select an appropriate license from the GitHub options list on repository creation. - Usually, the MIT license is appropriate.
SCRIPTS folder	- Goal: This folder contains all the source code for your project. - Include all the scripts you used. Try to name each script according to the order it needs to be executed to reproduce the results. - All script files should include header comments at the beginning of a script to provide information that anyone working with or executing the script should be aware of. **Throughout all your scripts, you should include copious comments explaining what each command or sequence of commands accomplishes and what the purpose is**.
DATA folder	- Goal: This folder contains all of the data for this project. - You should AT LEAST the data include the initial data, and the final data analyzed. If needed, the code in the SCRIPTS folder should be able to get you from the initial piece of data to the final one. - If your data fits in github, place all of it here. - If your data does not fit in GitHub use a single file explaining the process to obtain the dataset. - A Data Appendix file as a PDF, which will include text that you type, as well as tables, figures, and other descriptive statistics. This file should be organized in sections, with a section for **each** dataset analyzed. Each section should begin with a statement of what the unit of observation is--that is, it should explain what kind of object each row of the data file represents. After that, you should include a subsection for each variable in the analyzed dataset. More information: https://www.projecttier.org/tier-protocol/protocol-4-0/root/data/analysisdata/data-appendixfile
OUTPUT folder	- Goal: This folder contains all of the output generated by your project, e.g. figures, tables, etc. - Importantly, any information like tables and figures should be here. **Use informative names for your files**.
References	- All references should be listed at the end of the document - Use IEEE Documentation style (link)

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